

Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

End Semester Examination *SET I Synoptic*

November 2017

Max. Marks: 100

Class: TE

Course Code: CPC502

Name of the Course: Operating System

Duration: 180 Minutes

Semester: V

Branch: COMPUTER

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks	CO
Q 1 (a)	What is Operating System? Write three function of it? Definition: 2 marks 3 function: 3 marks	5	CO1
Q1 (b)	What is inode in Unix system ? Explanation 4 marks 1 mark diagram	5	CO6
Q1 (c)	What is role of Process Control Block? PCB content 2 marks. Its use 3 marks	5	CO1
Q1 (d)	What do you mean by system call? Discuss any one file system call giving suitable example? Definition 2 marks. One Example stating function name at user level and function name at kernel level 3 marks	5	CO1
Q2 (a)	Discuss different file allocation methods . 3 allocation methods with proper diagrams.	10	CO5
Q2(b)	Suppose that a disk drive has 200 cylinders 0 to 199. Assuming that disk head is moving towards 0 and the drive is currently serving the request at 53. consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 37, 122, 14, 124, 65, 67 in that order . Find the average seek time using SSTF, SCAN algorithms. 2 methods with proper diagrams 5 marks each, SSTF: 29.5, SCAN: 29.75	10	CO5
Q3(a)	Discuss hardware approach to mutual exclusion. Definition of mutual exclusion : 2 marks, any one H/w approach 8	10	CO3

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Q4 (a)	Explain Demand Paging ? What are causes of thrashing? How to overcome thrashing?	10	CO4
	Explanation : 3 marks, thrashing with graph 3 marks solution 4 marks OR		
Q4(a-1)	Consider a computer system with 24 bit Physical address and 33 bit logical address and page size of two kilobytes. find no. of bits for offset, no. of pages and no. of page frames? Offset 2 marks , 1.5 each for no. of pages and no. of frames Offset:11 bits,no. of frames: 2^{13} ,no of pages : 2^{22}	5	CO4
Q4(a-2)	Let the page fault service time be 8 milliseconds(ms) in a computer with average memory access time being 200 nanoseconds (ns). If one page fault is generated every 1000 memory accesses, what is the effective access time for memory? Formula:1 marks, conversion :2 marks : EAT value 2 marks $EAT = (1-p) * \text{memory access time} + p * (\text{page fault overhead} + \text{swap page out} + \text{swap page in})$ $8.1998 \sim 8.2 \text{ microseconds}$	5	CO4
Q4 (b)	Explain Hardware support for paging with TLB ? Diagram with explanation 10 marks	10	CO4
Q4(b)	OR Calculate no of faults for following reference string using LRU and Optimal page replacement policy .Assume no of frames =3 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1 Diagram showing frame contents at every iteration 10 marks and no of page fault 1 marks & no. of page faults , LRU : Fault=12,Optimal : faults =9	10	CO4
Q5 (a)	Explain Unix file system. Unix File system layout 4 marks explanation of each components 6 marks	10	CO6
Q5(a)		10	CO6

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	OR Explain process scheduling in Unix system. Type of process scheduling in unix system Priority based RR :2 marks , nice() call: 2 marks , fair share scheduler :2 marks diagrams 4 marks																										
Q5 (b)	Consider the following set of processes. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Process name</th><th>Arrival time</th><th>Processing time</th><th>Priority</th></tr> </thead> <tbody> <tr> <td>A</td><td>0</td><td>3</td><td>1</td></tr> <tr> <td>B</td><td>2</td><td>6</td><td>4</td></tr> <tr> <td>C</td><td>4</td><td>4</td><td>5</td></tr> <tr> <td>D</td><td>6</td><td>5</td><td>2</td></tr> <tr> <td>E</td><td>8</td><td>2</td><td>3</td></tr> </tbody> </table> Find average turnaround time using Non Preemptive Priority and RR quantum 1. Draw Gantt chart. Calculation and gantt chart 5 marks each. Non Preemptive Priority: $42/5=8.4$ and RR =10.8	Process name	Arrival time	Processing time	Priority	A	0	3	1	B	2	6	4	C	4	4	5	D	6	5	2	E	8	2	3	10	CO2
Process name	Arrival time	Processing time	Priority																								
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